

FAME CONSULTANT PTY LTD

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Consulting Structural Engineers
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31st March, 2026
Ref.: 7918/2026/FCSTR/R01

TO WHOM SO EVER IT MAY CONCERN

Re: Proposed residential development at
1, Arjez Place, Marayong, NSW

I have been engaged to undertake structural design of the proposed residential development at above property – double storey dwelling and granny flat.

Structural design of the proposed development at above mentioned property has been done to fulfill requirements of following:

- (1) Structural requirements of NCC Vol.2; BCA 2022; Housing Provision standards 2022
- (2) AS 2870-2011: Residential Slabs and Footing for **Class “H2” Site**
- (3) AS 3600-2018: Concrete Structures
- (4) AS 1684.2-2021: Residential timber-framed construction (non-cyclonic area)
- (5) AS 4100-2020: Steel Structures
- (6) AS 1170.0-2002: Structural design actions —General Principles
- (7) AS 1170.1-2002 (R2016): Structural design actions — Permanent, imposed and other actions
- (8) AS 1170.2-2021: Structural design actions — Wind actions
- (9) Lot classification report CG26-0486-A Rev 0; 24/03/2026

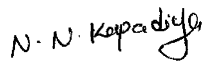
I am professionally qualified engineer and as such can certify that the performance of the design complies with the above-mentioned standards and detailed on below structural drawings:

- 1) SD-01 to SD-06; Revision-01; dated – 31/03/2026

Pier and beam type construction has been adopted for the proposed development.

This certification does not relieve all stake holders involved in the project of their professional responsibilities.

Please feel free to contact for further details.



Nimanshu N. Kapadiya
BE, MIE (Aust), CPEng, NER, RPEQ
Member No: 3268731

PROPOSED RESIDENTIAL DEVELOPMENT AT
1, ARJEZ PLACE, MARAYOUNG, NSW

NOTES:

GENERAL:

- THESE DRAWINGS TO BE READ IN CONJUNCTION WITH ARCHITECTURAL, STORM WATER AND ALL OTHER CONSULTANTS' DRAWINGS AND SPECIFICATIONS.
- DIMENSIONS SHALL NOT BE OBTAINED BY SCALING STRUCTURAL DRAWINGS.
- SETTING OUT DIMENSIONS SHOWN ON THE DRAWINGS SHALL BE VERIFIED BY THE BUILDER.
- DURING CONSTRUCTION, STRUCTURE SHALL BE MAINTAINED IN STABLE CONDITION AND NO PART SHALL BE OVER STRESSED.
- ALL WORKMANSHIP AND MATERIAL SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF SA CODES AND ORDINANCES OF RELEVANT AUTHORITY.
- ALL DIMENSIONS ARE IN MILLIMETERS AND LEVELS ARE IN METER U.O.N.

FOOTING:

- FOOTINGS ARE TO BE FOUNDED ON ORIGINAL UNDISTURBED GROUND HAVING SAFE BEARING CAPACITY OF 250kPa. SAFE BEARING CAPACITY TO BE VERIFIED.
- THE SITE HAS BEEN CONSIDERED AS CLASS 'P' /H2. IN ACCORDANCE WITH AS: 2870-2011 REFER LOT CLASSIFICATION REPORT CG26-0486-A Rev 0; 24/03/2026.
- CARRY OUT TERMITE PROTECTION AS PER THE REQUIREMENTS OF STANDARDS AND NATIONAL CONSTRUCTION CODE.
- PLACE 0.2mm THICK FORTECON MEMBRANE TO UNDERSIDE OF SLAB AND BEAM. LAP ALL JOINTS 300mm MIN. AND SEAL WITH 50mm WIDE TAPE TAPE ALL PENETRATIONS TO ENSURE MAINTENANCE OF BARRIER.

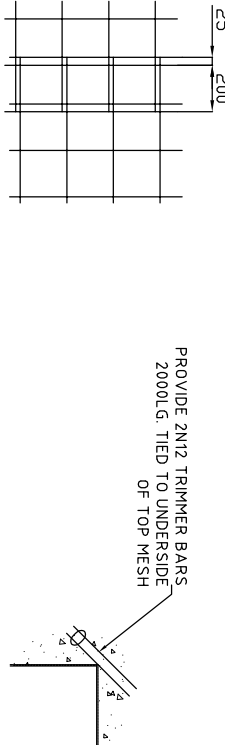
CONCRETE:

- ALL WORKMANSHIP AND MATERIAL SHALL BE IN ACCORDANCE WITH AS: 3600 CURRENT EDITION WITH ALL AMENDMENTS.
- MIN. CONCRETE QUALITY (ACCORDANCE TO AS: 3600)

ELEMENT	SLUMP	AGGREGATE (MAX. SIZE)	f'c CEMENT TYPE
FOOTING & PIERS	80	20	25MPa GP
WAFFLE SLAB	80	20	25MPa GP

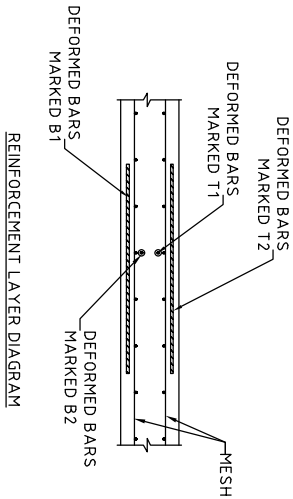
- CLEAR CONCRETE COVER TO REINFORCEMENT SHALL BE AS FOLLOWS UNLESS OTHERWISE SHOWN:

ELEMENT	CAST AGAINST FORMWORK	EXPOSED TO GROUND	CAST AGAINST GROUND	EXPOSURE CLASS
COLUMN & PEDESTAL	50	50	50	A2
BEAMS	50	50	50	A2
FOOTING	50	50	50	A2
SLABS & WALLS	25	45	50	A2

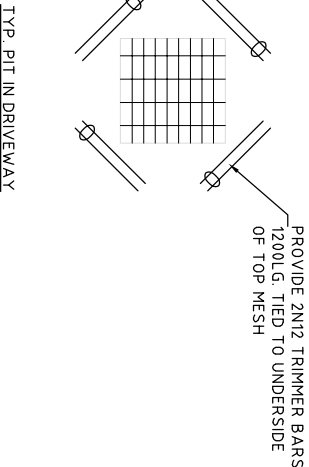


SLAB MESH LAPPING DETAILS

TYP. TRIMMER AT RE-ENTRANT CORNER



REINFORCEMENT LAYER DIAGRAM

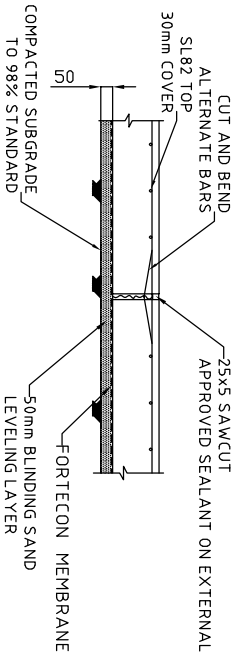
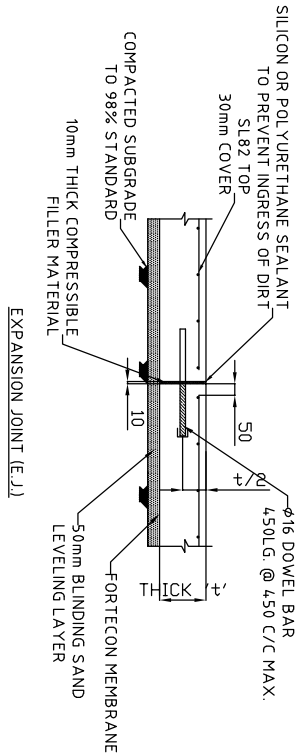


TYP. PIT IN DRIVEWAY

- SIZES OF CONCRETE ELEMENTS DO NOT INCLUDE THICKNESS OF APPLIED FINISHES.
- CONSTRUCTION JOINTS WHERE NOT SHOWN SHALL BE LOCATED TO THE APPROVAL OF ENGINEER.
- BEAM DEPTHS ARE WRITTEN FIRST AND INCLUDE SLAB THICKNESS. NO HOLES OR CHASES OTHER THAN THOSE SHOWN ON THE STRUCTURAL DRAWING SHALL BE MADE IN CONCRETE MEMBERS WITHOUT PRIOR APPROVAL OF THE ENGINEER.
- REINFORCEMENT SHOWN DIAGRAMMATICALLY. IT IS NOT NECESSARILY SHOWN IN TRUE PROJECTION.
- SPLICES IN REINFORCEMENT SHALL BE MADE ONLY IN THE POSITION SHOWN. THE WRITTEN APPROVAL OF THE ENGINEER SHALL BE OBTAINED FOR ANY OTHER SPLICES.
- WELDING OF REINFORCEMENT WILL NOT BE PERMITTED UNLESS SHOWN ON THE STRUCTURAL DRAWINGS.
- REINFORCEMENT SYMBOLS:

- N: HOT ROLLED DEFORMED BAR TO AS: 4671; GRADE 500 N
- S: HOT ROLLED DEFORMED BAR TO AS: 4671; GRADE 230 S
- R: PLAIN ROUND BARS TO AS: 4671; GRADE 230 R
- SL: WELDED WIRE FABRIC IN ACCORDANCE WITH AS: 4671
- THE NUMBER FOLLOWING THE BAR SYMBOL IS THE NOMINAL BAR DIA IN mm.

- LAP BARS TO BE 50 TIMES BAR DIAMETERS, U.O.N.
- ALL REINFORCEMENT SHALL BE PROVIDED WITH SUFFICIENT CHAIRS, SUPPORTS AND ADDITIONAL TIE BARS WHERE NECESSARY SO THAT CORRECT POSITION IS MAINTAINED DURING CONCRETING. CHAIRS TO BE SPACED AT 900mm MAXIMUM CENTRES EACH WAY.
- WHERE CONCRETE BEARS ON BRICKWORK IT SHALL BE SEPARATED THEREFROM BY TWO LAYERS OF MALTHOID OR GALVANIZED SHEET STEEL.
- CONCRETE IS TO BE COMPACTED USING HIGH FREQUENCY VIBRATORS.
- BRICK WORK SHALL NOT BE BUILT OVER SUSPENDED SLAB UNTIL ALL FORMWORK AND PROPS ARE REMOVED.
- CONCRETE TO BE MOIST CURED FOR 7 DAYS AFTER POURING USING HYDROCARBON RESIN TO AS: 3799
- FLOOR SLABS ON GROUND SHALL BE POURED IN ALTERNATE PANELS BETWEEN JOINTS.
- REINFORCEMENT CROSSING PENETRATION SHALL BE DISPLACED AS NECESSARY NO REINFORCEMENT SHALL BE CUT WITHOUT PRIOR WRITTEN APPROVAL OF PROJECT ENGINEER.
- FORM WORK SHALL BE DESIGNED AND CONSTRUCTED IN ACCORDANCE WITH AS: 3600 AND AS: 3610
- PROVIDE PROPS TO ALL STRUCTURAL ELEMENTS AS PER SITE REQUIREMENT AND IN ACCORDANCE WITH INDUSTRY STANDARDS.



SAW CUT TO BE MADE WITHIN 36 HOURS OF PLACEMENT OF SLAB. SLAB TO BE CONTINUOUSLY WET UNTIL SAWCUT MADE AND FOR 5 DAYS CONTINUOUSLY AFTER CUTTING.

BRICKWORK & BLOCKWORK:

- ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS: 3700.
- BRICKWORK SHALL BE SEPARATED FROM CONCRETE AT ALL VERTICAL FACES BY 10mm THICK EXPANSION JOINTS.
- STRENGTHS OF BRICKS, CLASS OF BLOCKS AND TYPE OF MORTAR SHALL BE AS FOLLOWS:

ELEMENT	STRENGTH OR CLASS	MORTAR TYPE
BRICKS	f'uc = 15MPa f'm = 6.67	M3 (1:1:6)

- BRICK WALLS SHALL HAVE VERTICAL EXPANSION JOINTS, WITH 10mm FOAM STRIP, AT MAXIMUM 5m CENTRES AND WITHIN 2m OF CORNERS.

STRUCTURAL STEEL:

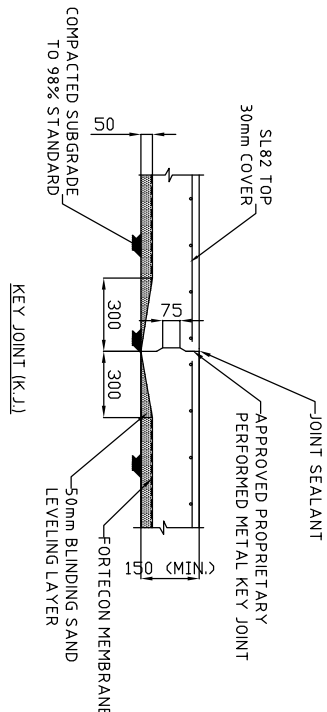
- ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS: 4700 AND AS: 1554.
- CONNECTION SHALL BE PROVIDED TO CARRY THE REACTIONS SHOWN OTHERWISE DETAILED.
- THE CONTRACTOR SHALL PREPARE WORKSHOP DRAWINGS AND SHALL SUBMIT TWO COPIES OF EACH DRAWING FOR APPROVAL. FABRICATION SHALL NOT COMMENCE UNTIL APPROVAL HAS BEEN RECEIVED.
- WELDS SHALL BE 6mm CONTINUOUS FILLET. ALL BOLTS TO BE ϕ 20 BOLTS AND ALL GUSSET PLATES TO BE 10mm THICK UNLESS OTHERWISE NOTED.
- HIGH STRENGTH FRICTION GRIP BOLTS AND HIGH STRENGTH BEARING BOLTS, NUTS AND HARDENED WASHERS ARE TO COMPLY WITH AS: 1252. METHOD OF TIGHTENING IS TO BE IN ACCORDANCE WITH AS: 1511 PART TURN OF NUT METHOD.
- ALL CLEATS AND DRILLING FOR FIXING OF TIMBER MEMBERS, ETC TO BE PROVIDED BY FABRICATOR.

REINFORCEMENT:

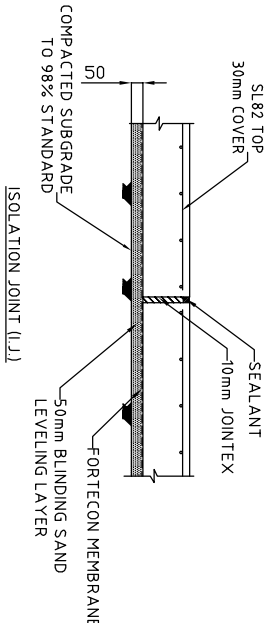
- ALL REINFORCEMENT SHOWN IS TO BE ADDITIONAL TO STANDARD REINFORCEMENT.
- REFER TO SECTION ON THIS DRAWING FOR STANDARD REINFORCEMENT.
- DISTRIBUTION REINFORCEMENT TO BE N16-300 LAPPED U.O.N.
- STRUCTURAL ENGINEER TO INSPECT ALL REINFORCEMENT PRIOR TO POURING CONCRETE

TIMBER WORK:

- ALL WORKMANSHIP AND MATERIAL SHALL BE IN ACCORDANCE WITH AS: 1684.
- AS: 1684 SHALL BE APPLIED TO DOMESTIC CONSTRUCTION IN SHELTERED LOCATION
- SOFTWOOD TO BE MINIMUM OF F7 SEASONED AND HARDWOOD TO BE MINIMUM F17 SEASONED U.O.N
- EXTERNAL TIMBER SHALL BE EITHER HARDWOOD DURABILITY CLASS 1 OR 2 AS PER AS: 1720 OR IMPREGNATED PINE GRADE F7, PRESSURE TREATED TO AS: 1604 AND PRE DRILLED PRIOR TO USE. SUPPLEMENTARY TREATMENT SHALL BE APPLIED TO ALL CUT SURFACES.
- ALL BOLTS IN TIMBER CONSTRUCTION SHALL BE MINIMUM M16, GRADE 4.6 U.O.N. WASHERS UNDER HEADS AND NUTS SHALL BE AT LEAST 2.5 TIMES THE BOLT DIAMETER.
- ALL TIMBER JOINTS AND NOTCHES SHALL BE A MINIMUM OF 100mm AWAY FROM LOOSE KNOTS, SEVERE SLOPING GRAIN, GUM VEINS OR OTHER MINOR DEFECTS.



KEY JOINT TO BE APPROVED PREFORMED METAL KEY, SECURELY STAKED INTO SUBGRADE



R01	ISSUED FOR ENGINEERING CERTIFICATE	31.03.2026	ARJUN
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TRUE NORTH
SCALE 1:30@A3

DRAWING NUMBER: SD-01
DRAWING TITLE: CONSTRUCTION NOTES

REV. NO.	PROJECT
01	PROPOSED RESIDENTIAL DEVELOPMENT AT 1, ARJEZ PLACE, MARAYOUNG, NSW

SOIL CLASS : "H2"

SUMMARY:-

SITE CLASSIFICATION: P
SOIL CLASSIFICATION: H2

DETAILS:-

SLAB THICKNESS : 100mm
PIERS TYPE : CONCRETE PIERS
SIZE OF PIERS : Ø400mm CONCRETE BORED PIERS
BEAM CROSS SECTION : REEFER SD-03 & SD-04

POD SIZE:-

MAIN SLAB	: 300mm
GARAGE	: 225mm
PORCH/ALFRESCO	: 225mm

CONCRETE GRADE:-

SLAB : 25MPa
PIERS : 25MPa
FOOTING : 25MPa

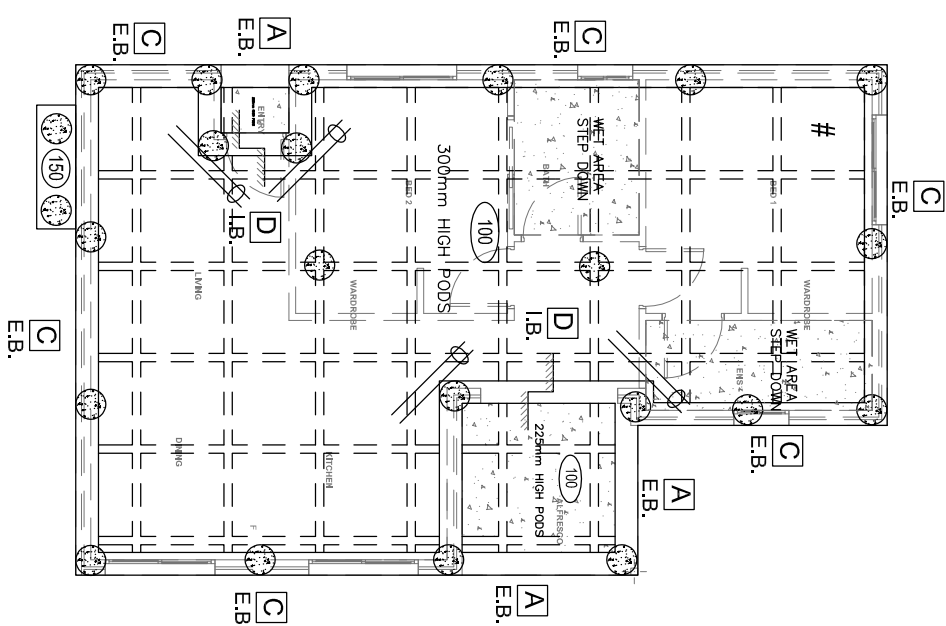
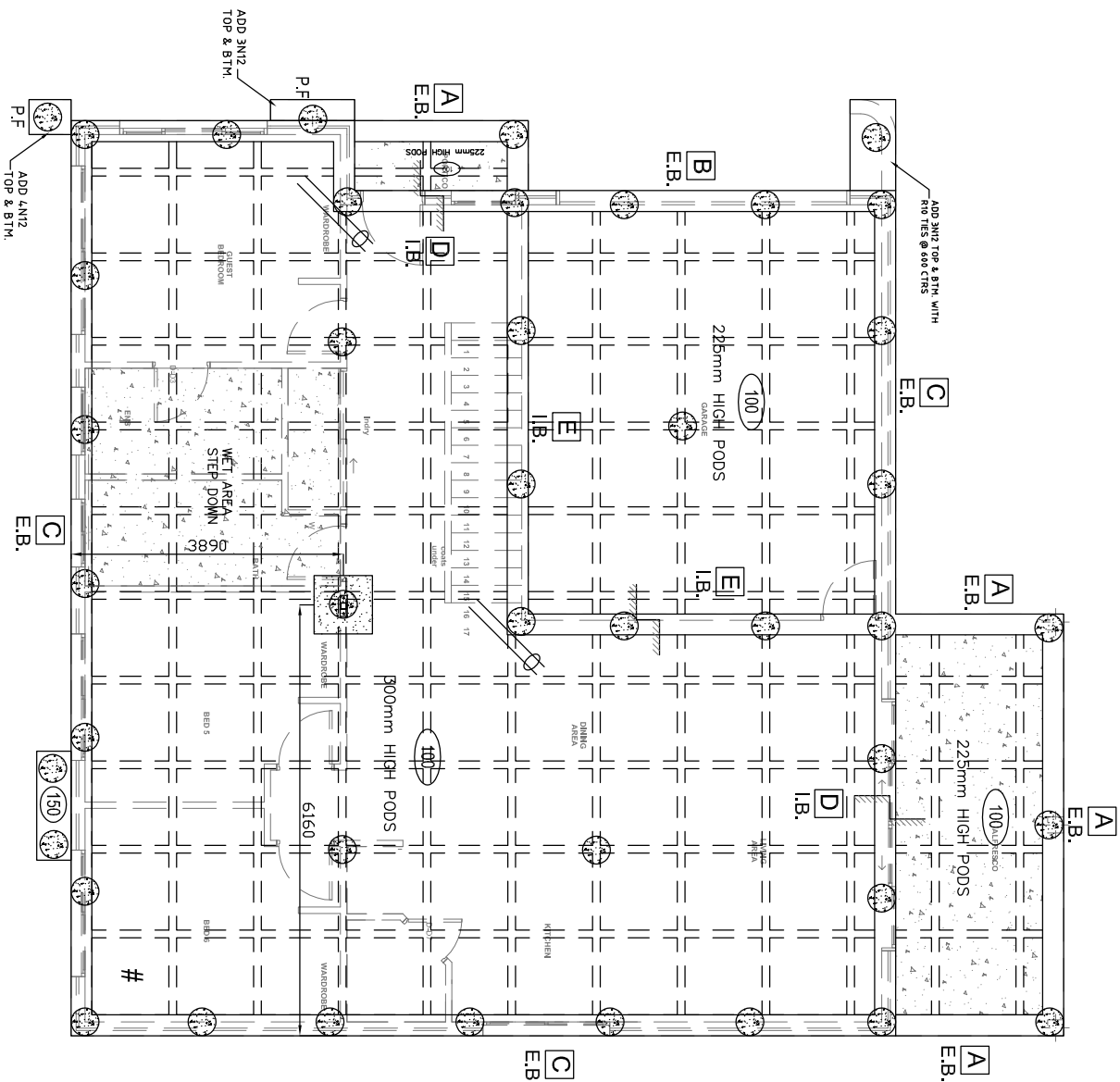
NOTES:-

MIN. PIERS DEPTH : 2m
IF PIERS DEPTH EXCEED 3m, CONSULT OUR OFFICE.

SALINITY NOTES:-

- 1) SALINITY CLASS : A2
- 2) PLACE HIGH IMPACT RESISTANCE MIN. 0.2mm THICK WATERPROOFING MEMBRANE UNDER SLAB/FOOTING.
- 3) PROVIDE 50mm LAYER OF SAND UNDER THE SLAB.
- 4) REFER TABLE FOR COVER TO REINFORCEMENT.
- 5) EXPOSURE CLASS MASONRY UNITS MUST BE USED UNDER THE DAMP PROOF COURSE INCLUDING STRIP FOOTING.

- REFER ARCHTECT PLANS FOR STEP DOWNS AND LEVELS.
PROVIDE Z-BARS TO ALL DROP EDGE BEAM AS PER SD-03.
IF HIT SHALE DURING DRILLING, ALL PIERS MUST BE DRILLED TO SHALE.
BRACING, ROOF WALL, FLOOR FRAMES DESIGNED BY FRAME COMPANY.



LEGENDS:-

- | | |
|---|---|
|  | DENOTES 500*500 CUT IN WAFFLE POD FOR STEEL POSTS WITH MESH AT BOTTOM |
|  | DENOTES WAFFLE SLAB STEP-DOWN |
|  | DENOTES SLAB THICKNESS |
|  | DENOTES EDGE BEAM |
|  | DENOTES START POD |
|  | DENOTES INTERNAL BEAM |
|  | DENOTES 1100 MM X 1100 MM X 225 MM WAFFLE PODS |
|  | DENOTES 2N12 - 2000 LONG EXTRA STEEL AT TOP TIED UNDERSIDE OF SLAB MESH |
|  | DENOTES Ø400 MASS CONCRETE PIER |



R01	ISSUED FOR ENGINEERING CERTIFICATE	31.03.2026	ARJUN
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TRUE NORTH

TRUE NORTH

SD-02

REV. NO.

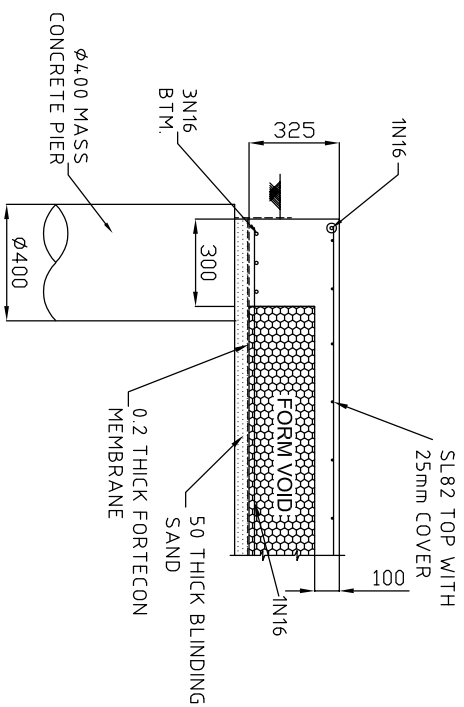
PROJECT

DRAWING TITLE:

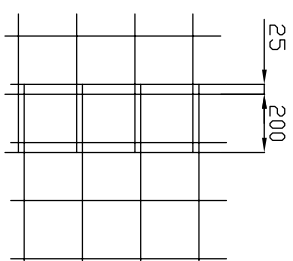
LAYOUT FOR WAFFLE POD SLAB AND FOOTING BEAM

01

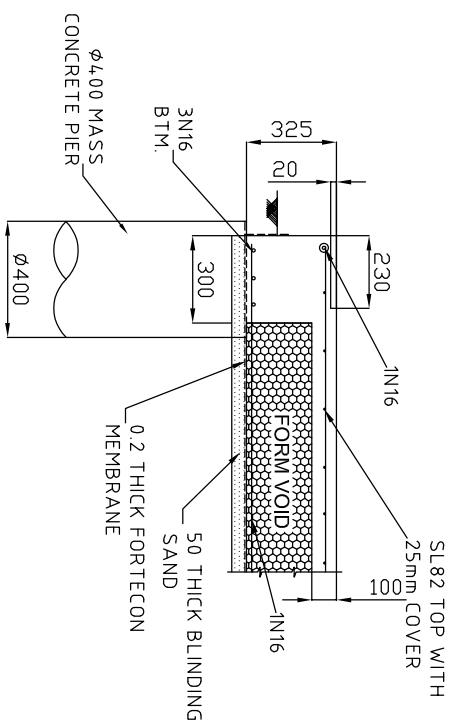
PROPOSED RESIDENTIAL DEVELOPMENT AT
1, ARJEZ PLACE, MARAYOUNG, NSW



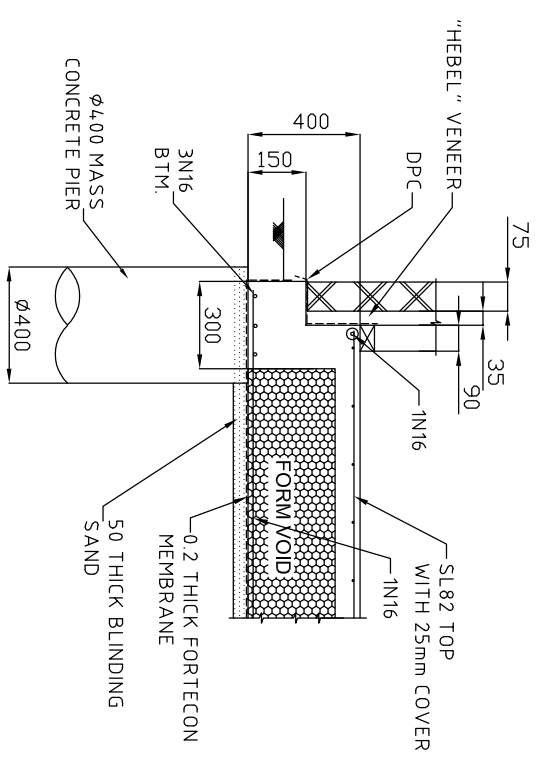
TYPE 'A' FOOTING



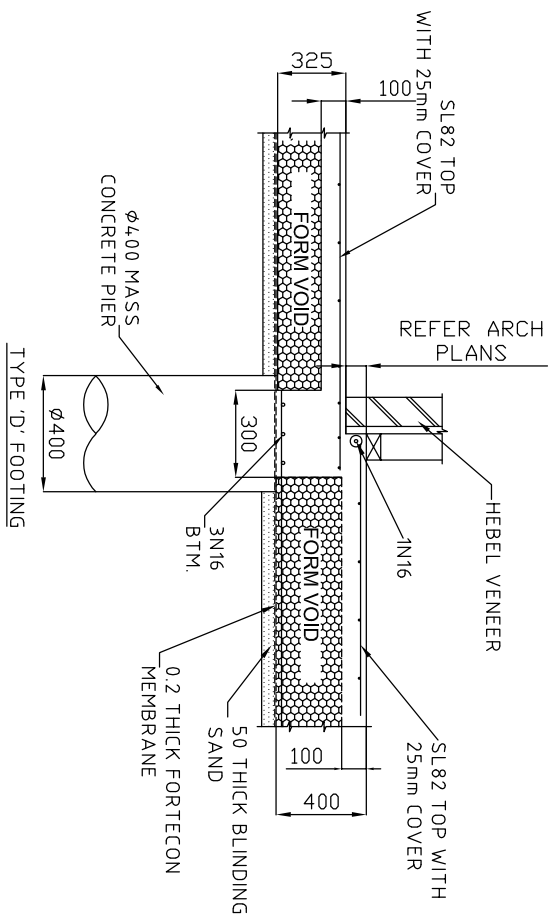
SLAB MESH LAPPING DETAILS



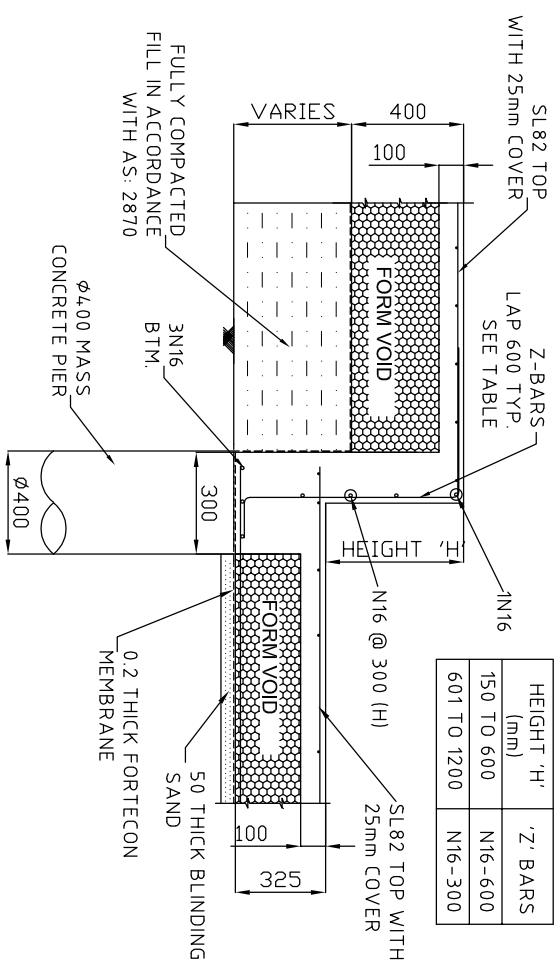
TYPE 'B' FOOTING



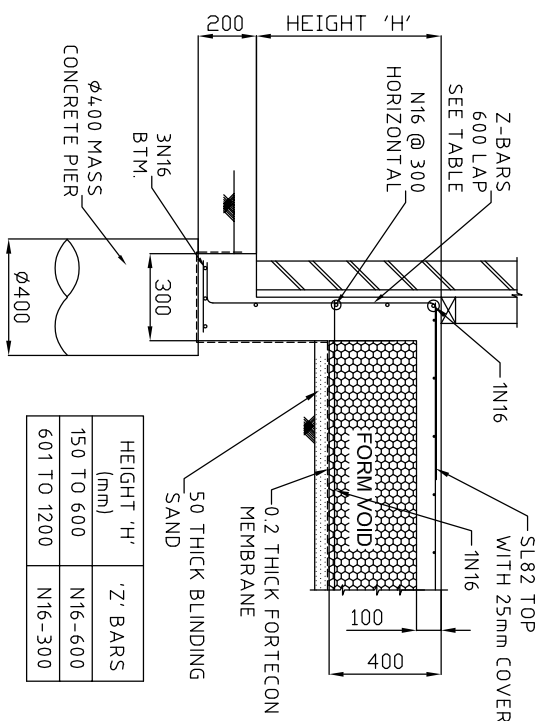
REBATE DETAILS ARE FOR GUIDANCE PURPOSE ONLY.
REFER "HEBEL" MANUFACTURER'S SPECIFICATIONS.



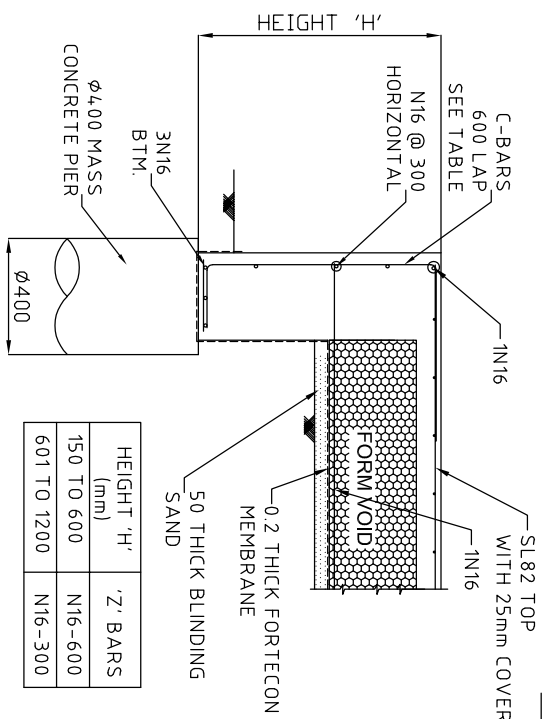
TYPE 'D' FOOTING



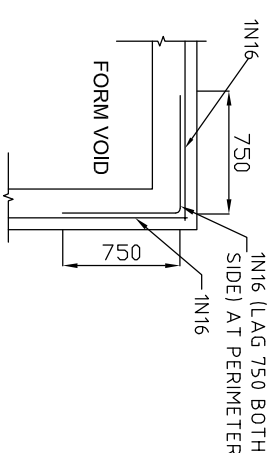
TYPE 'E' FOOTING



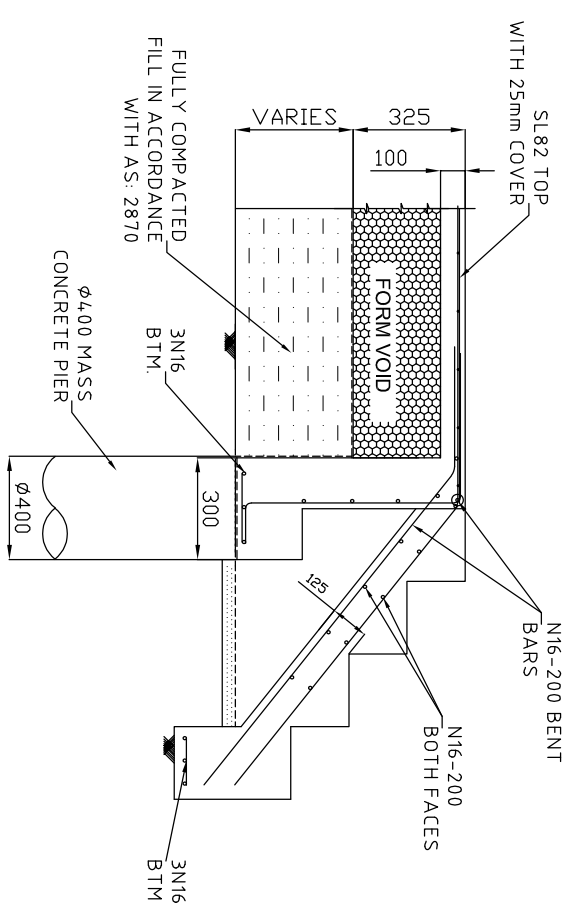
ALTERNATE EDGE BEAM DETAILS



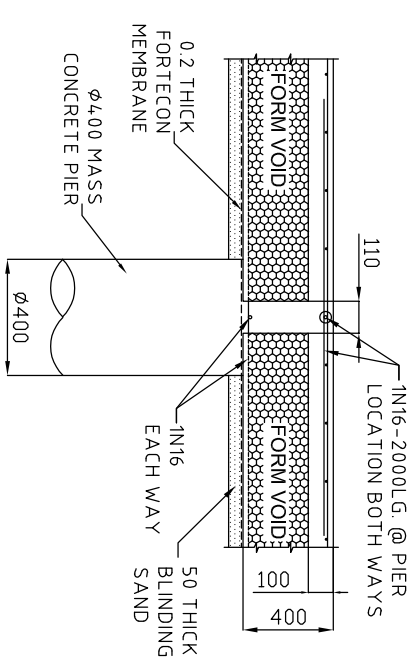
ALTERNATE EDGE BEAM DETAILS



PERIMETER LAPPING DETAILS



TYPE STEPS DETAILS

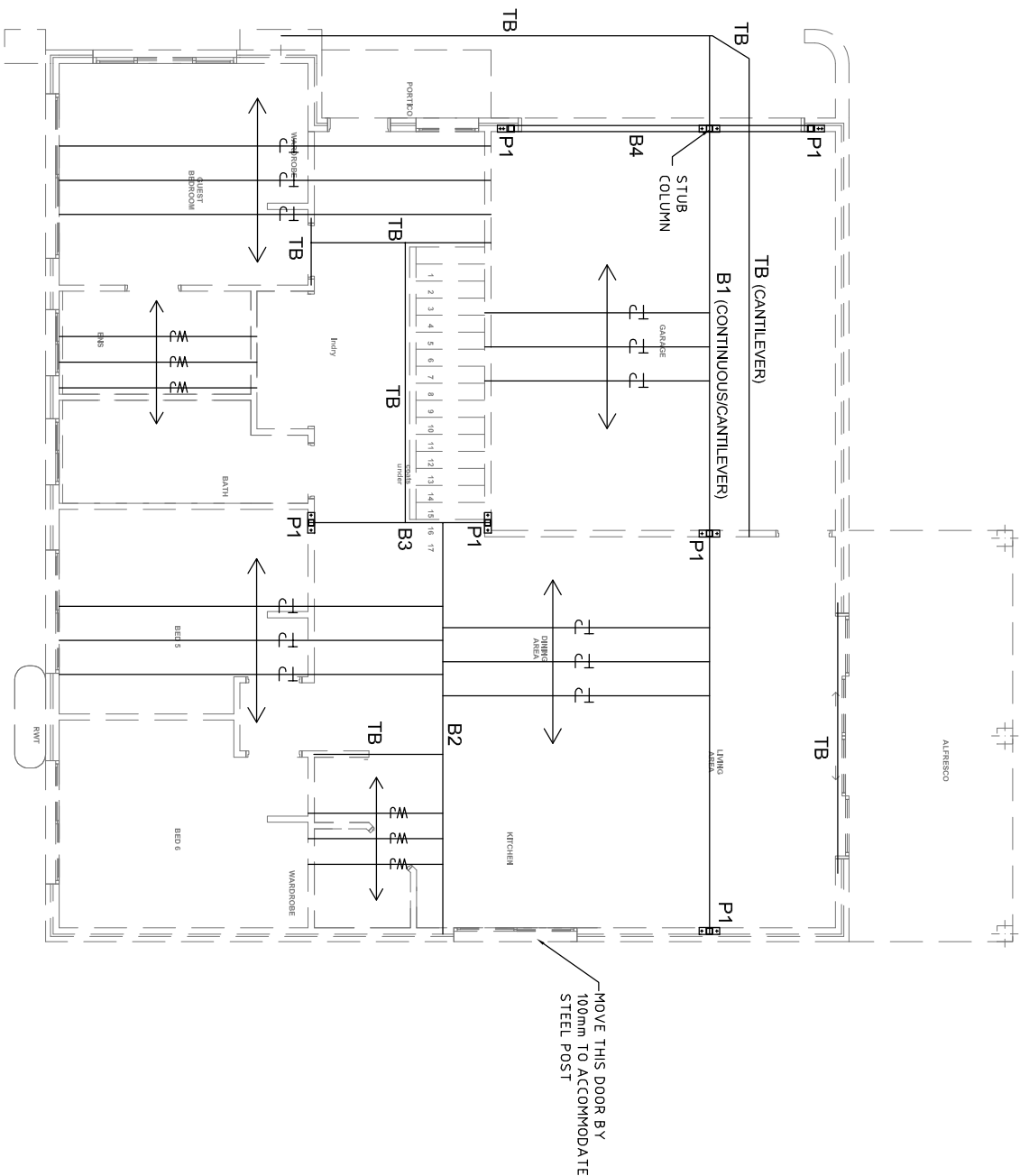


INTERNAL RIB DETAILS



1:100@A3

LAYOUT FOR FIRST FLOOR



MEMBER	SIZE	AS	GRADE/ty
P1	SHS: 89x89x6	AS1163	C350
TJ	TIMBER JOIST DESIGNED BY FRAMING COMPANY	AS1684	
TB	TIMBER BEAM DESIGNED BY FRAMING COMPANY	AS1684	
B1	310UB46	AS3679	300
B2	310UB46	AS3679	300
B3	300PFC	AS3679	300
B4	380PFC	AS3679	300

STEEL MEMBER SCHEDULE

- 1 REFER SHELS-D-01 FOR RELEVANT STANDARD DETAILS.
- 2 ARTICULATION JOINT TO BE CONSTRUCTED FOR FULL HEIGHT OF WALL. LOCATION TO BE CONFIRMED WITH ARCHITECT & BUILDER.
- 3 ALL STEEL SHALL COMPLY RELEVANT AUSTRALIAN STANDARDS
4. CONSULT ENGINEER, IF THERE IS CHANGE IN TIMBER JOIST POSITION
- 5 ALL STEEL WORK TO BE PAINTED AS PER ABCB HOUSING PROVISIONS STANDARD CLAUSE 6.3.9 (NCC2022) SPECIFICATION. BUILDER TO CONFIRM.
- 6 ALL FASTENERS TO BE HOT DIP GALVANISED AND OF 8.8 GRADE
- 7 ALL SITE WELDING TO BE CARRIED OUT BY QUALIFIED WELDER.
- 8 LIGHT WEIGHT STRUCTURAL CLADDING AT FIRST FLOOR COMPLIANCE TO BE CHECKED BY OTHERS.

LEGENDS

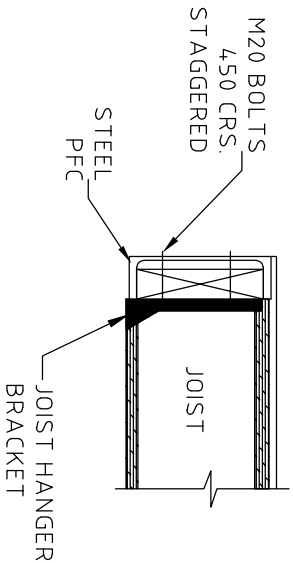
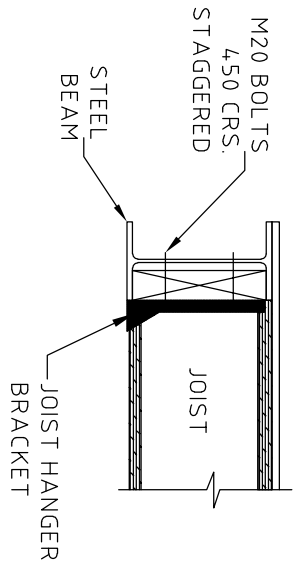
- | | |
|-------------|---------------------------|
| P1,P2,.. | DENOTES STEEL POST |
| B1,B2,.. | DENOTES STEEL BEAM/LINTEL |
| TB1, TB2,.. | DENOTES TIMBER BEAM |
| T.J. | DENOTES TIMBER JOIST |

R01	ISSUED FOR ENGINEERING CERTIFICATE	31.03.2028	ARJUN
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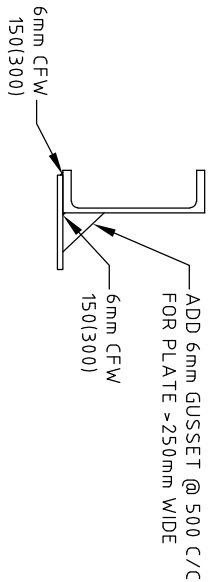
CONSULTING STRUCTURAL ENGINEERS
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TRUE NORTH	DRAWING NUMBER: SD-05	REV. NO.	PROJECT
	DRAWING TITLE: STEEL BEAM LAYOUT	01	PROPOSED RESIDENTIAL DEVELOPMENT AT 1, ARLEZ PLACE, KARAYOUNG, NSW
SCALE	1:100@A3		

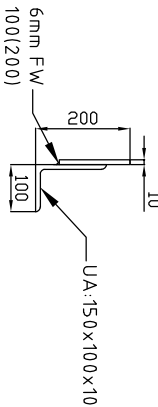
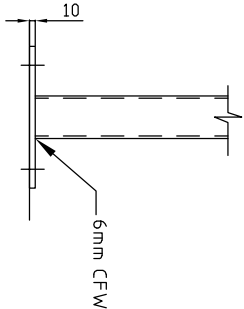


JOIST TO BEAM
CONNECTION DETAILS

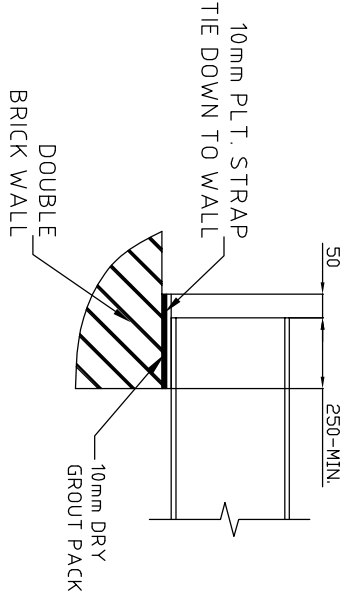
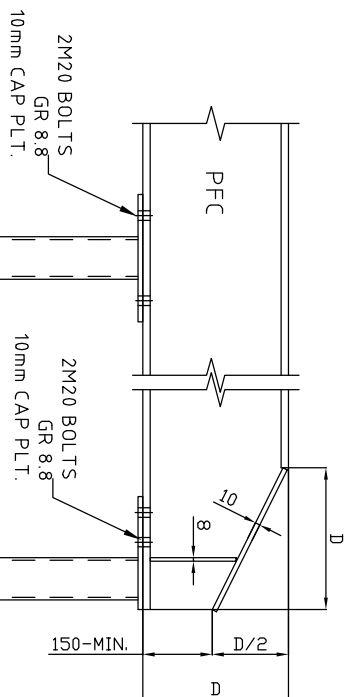
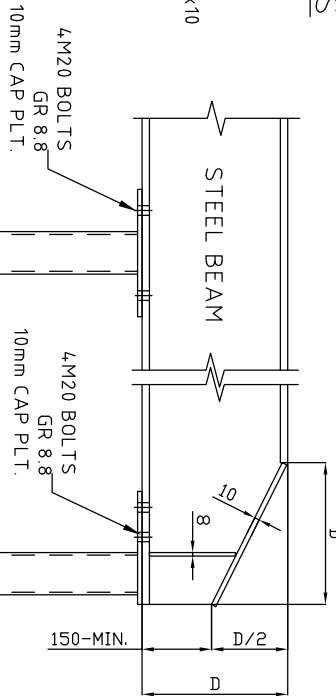
JOIST TO PFC
CONNECTION DETAILS



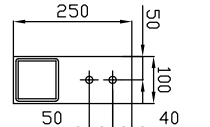
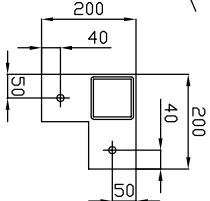
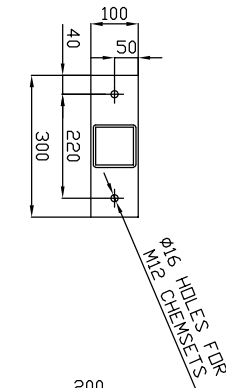
TYP. PFC-PLATE
CONNECTION DETAILS



DETAILS FOR 'L1'

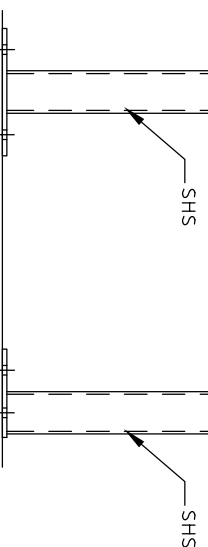
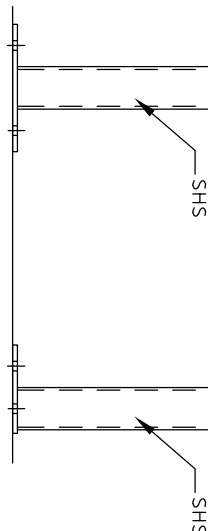


TYP. END BEARING
DETAILS

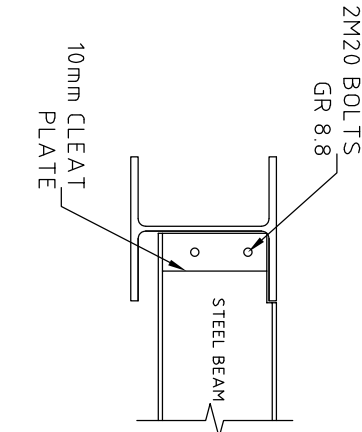


TYP. COLUMN BASE
DETAILS

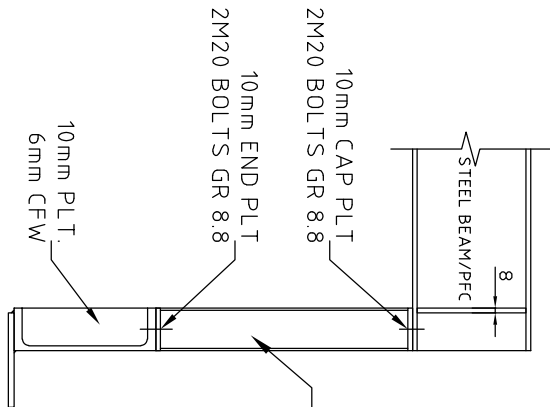
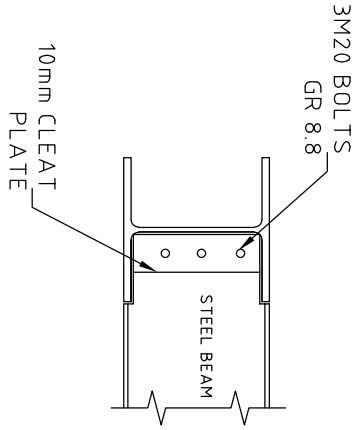
TYP. COLUMN TO STEEL
BEAM CONNECTION DETAILS



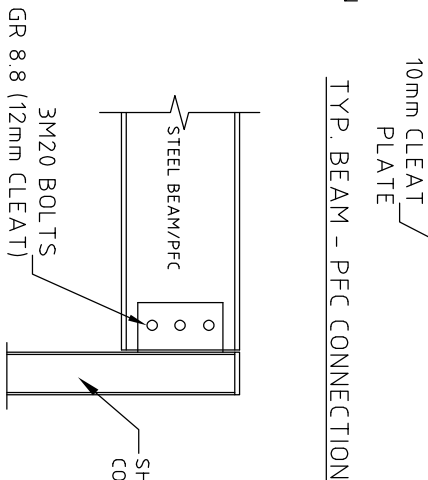
TYP. COLUMN TO PFC
CONNECTION DETAILS



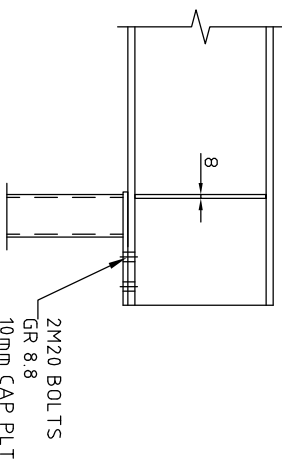
TYP. BEAM - BEAM
CONNECTION



TYP. BEAM - STUB
COLUMN CONNECTION



TYP. BEAM - PFC CONNECTION



TYP. BEAM - COLUMN CONNECTION

LINTEL SCHEDULE			
SPAN (mm)	EXTERNAL WALL (mm)	INTERNAL WALL (mm)	
UPTO 1200	FLAT 90x10	FLAT 90x10	
1201 TO 1500	EA. 100x100x10	EA. 100x100x6	
1501 TO 2400	UA 125x100x8	UA 125x100x8	
2401 TO 3000	UA 150x100x10	UA 150x100x10	